Layer Cut Experiment with CIFAR-100

SplitFedV2 (SFLV2) α=0.1

This experiment compares the accuracy and training time of early, mid, and late cut layers in **SplitFedV2 (SFLV2)** with ResNet-18 on CIFAR-100 α=0.1. For all three settings, the optimizer, learning rate, batch size, and data partitioning remains the same, only the cut layer changes.

Setup

- **Early cut:**
  - **Client:** Layer 1 → Layer 2 → Layer 3
  - **Server:** Layer 4 → Layer 5 → Layer 6 → AvgPool → fc
- **Mid cut:**
  - **Client:** Layer 1 → Layer 2 → Layer 3 → Layer 4 → Layer 5
  - **Server:** Layer 6 → AvgPool → fc
- **Late cut:**
  - **Client:** Layer 1 → Layer 2 → Layer 3 → Layer 4 → Layer 5 → Layer 6 → AvgPool
  - **Server:** fc

Comparison

Accuracy and Computation Time

Table 1: CIFAR-100, Dirichlet $\alpha = 0.1$: Test accuracy and computation time for different SFLV2 cut layers.

Model	Final Acc (%)	Best Acc (%)	Sum Acc	Total Time (min)
SFLV2 - LATE-Cut (6 + AvgPool)	**58.02**	**59.86**	**4653.86**	**56.5**
SFLV2 - MID-Cut (5)	47.41	48.67	3803.57	58.5
SFLV2 - EARLY-Cut (3)	36.86	38.87	2875.06	63.0

Mean Forgetting Loss

Table 2: CIFAR-100, Dirichlet $\alpha = 0.1$: Mean forgetting loss (lower is better) for different SFLV2 cut locations.

Model	Final Mean FgT Loss	Best Mean FgT Loss	Sum Mean FgT Loss
SFLV2 - MID-Cut (5)	-0.41	-1.45	-67.09
SFLV2 - EARLY-Cut (3)	-0.91	-1.41	-62.11
SFLV2 - LATE-Cut (6 + AvgPool)	**-0.49**	**-1.19**	**-77.33**

Accuracy

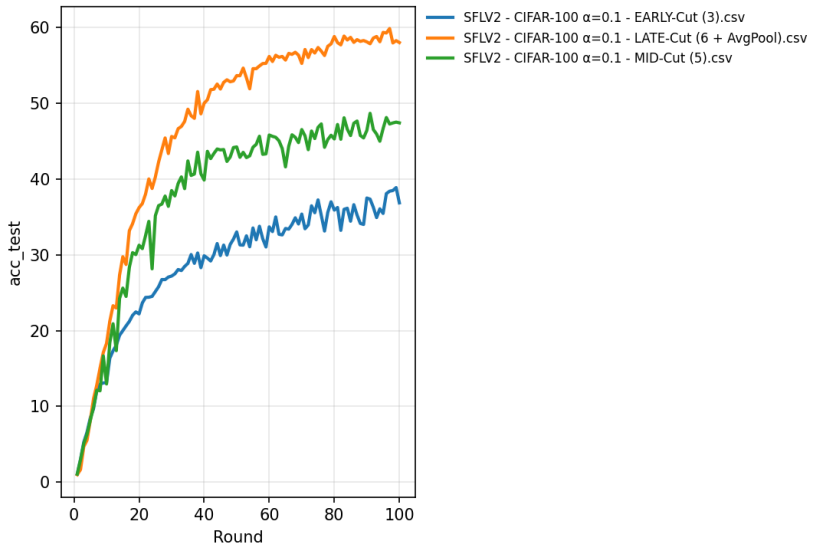

Figure 1: Accuracy Comparison between Different Cut Layers for SFLV2 on CIFAR-100

Computation Time

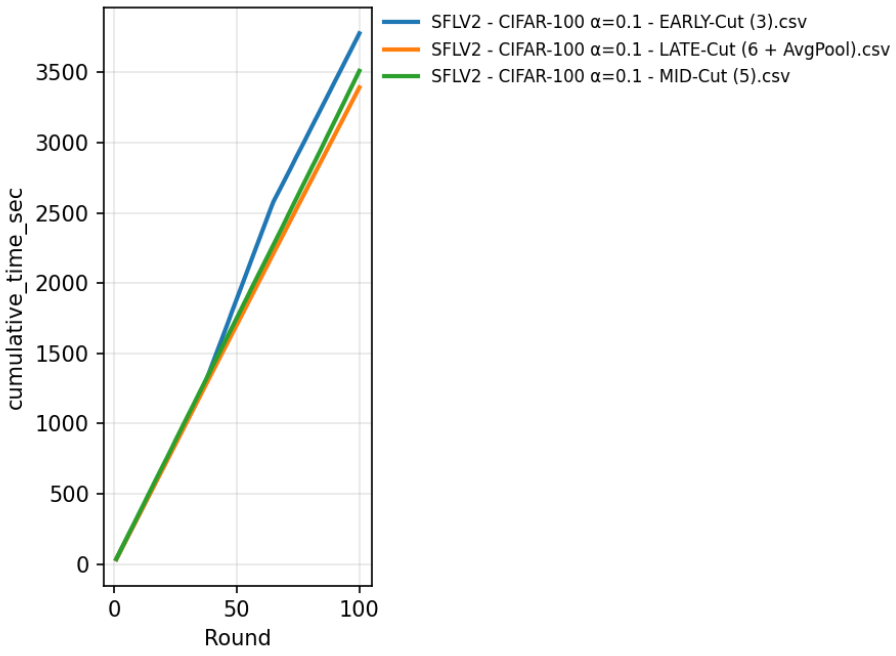

Figure 2: Training Time Comparison between Different Cut Layers for SFLV2 on CIFAR-100

Quantifying Forgetting

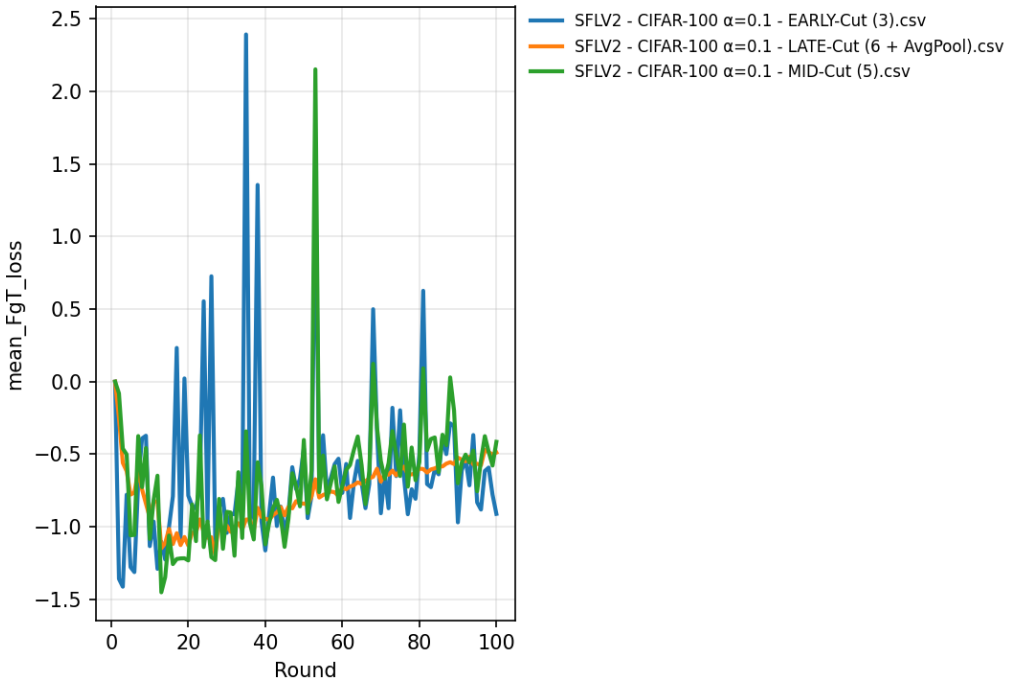

Figure 3: Quantifying Forgetting Comparison between Different Cut Layers for SFLV2 on CIFAR-100

Summary

After comparing the early, mid, and late cut layers, the late cut gave the bast results in terms of accuracy, computation time, and forgetting for Split Federated Learning Version 2 (SFLV2) on CIFAR-100 α=0.1. Therefore, the late cut will be used for further experiments and compares with other models.

Comparison Between Models

In **CIFAR-10** α=0.1 and α=0.5, The experiment setup for each model is shown in table below.;

Table 3: CIFAR-10, Dirichlet $\alpha = 0.1$ and α=0.5

Model	**Global Rounds**	**Clients**	**Local Epochs**	**LR**	**Warm-up**	**Note**
Centralized Baseline	100	10	0	0.01	0	-
FL (FedAvg)	100	10	5	0.01	0	-
FOT	100	10	5	0.01	5	-
SL	100	10	5	0.01	0	Cut at the end
SL + Projected	100	10	5	0.01	5	Rebuild every 3 rounds
SFLV2	100	10	5	0.01	0	Cut at the end **(best)**
SFLV2 + Projected	100	10	5	0.01	5	Cut at the end Rebuild every 3 rounds

CIFAR-10, α=0.1

Accuracy and Computation Time

Table 4: CIFAR-10, Dirichlet $\alpha = 0.1$: Test accuracy and computation time.

Model	**Final Acc (%)**	**Best Acc (%)**	**Sum Acc**	**Total Time (min)**
Centralized Baseline	**90.61**	**91.16**	**8621.22**	**17.7**
FL (FedAvg)	77.44	81.15	6757.69	75.8
FOT	65.01	65.01	5125.35	57.4
SL	36.11	40.32	3326.73	69.8
SL + Projected	46.87	46.87	3564.67	789.8
SFLV2	70.49	72.98	5420.62	57.2
SFLV2 + Projected	64.83	67.19	4079.53	70.3

Mean Forgetting Loss

Table 5: CIFAR-10, Dirichlet $\alpha = 0.1$: Mean forgetting loss **(lower is better)**.

Model	**Final Mean FgT Loss**	**Best Mean FgT Loss**	**Sum Mean FgT Loss**
Centralized Baseline	-0.12	-0.41	-14.75
FL (FedAvg)	-0.35	-1.00	-54.59
FOT	-0.01	-0.71	0.93
SL	-0.07	**-2.30**	-40.76
SL + Projected	**-1.05**	-1.33	-37.21
SFLV2	-0.56	-1.79	**-74.75**
SFLV2 + Projected	-0.57	-0.67	-41.63

Accuracy

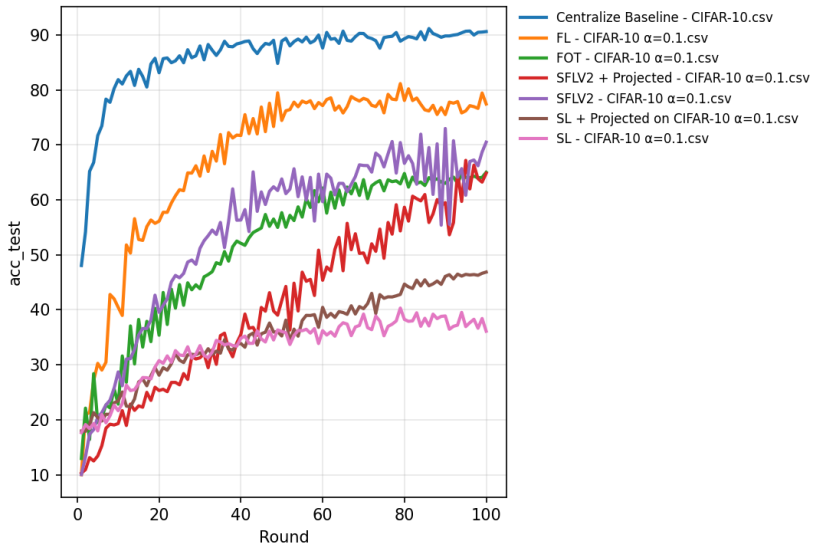

Figure 4: Accuracy Comparison between Tested Models

Computation Time

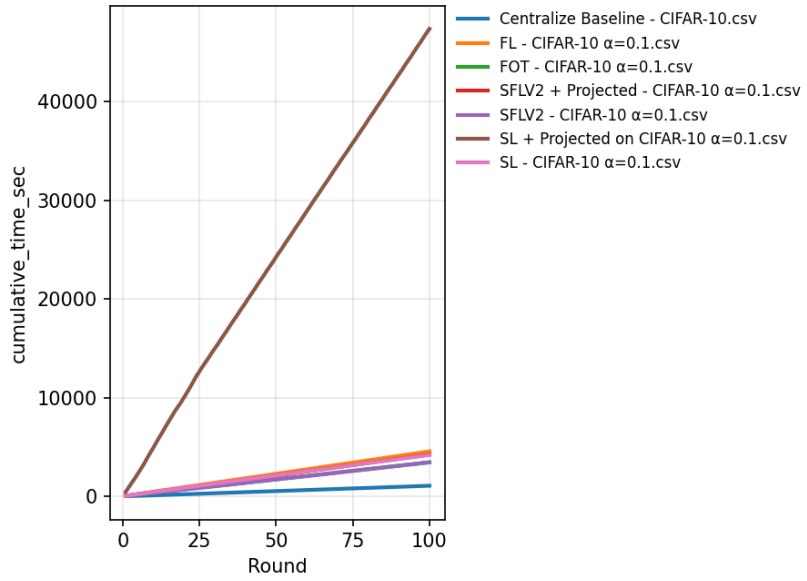

Figure 5: Training Time Comparison between Tested Models

Quantifying Forgetting

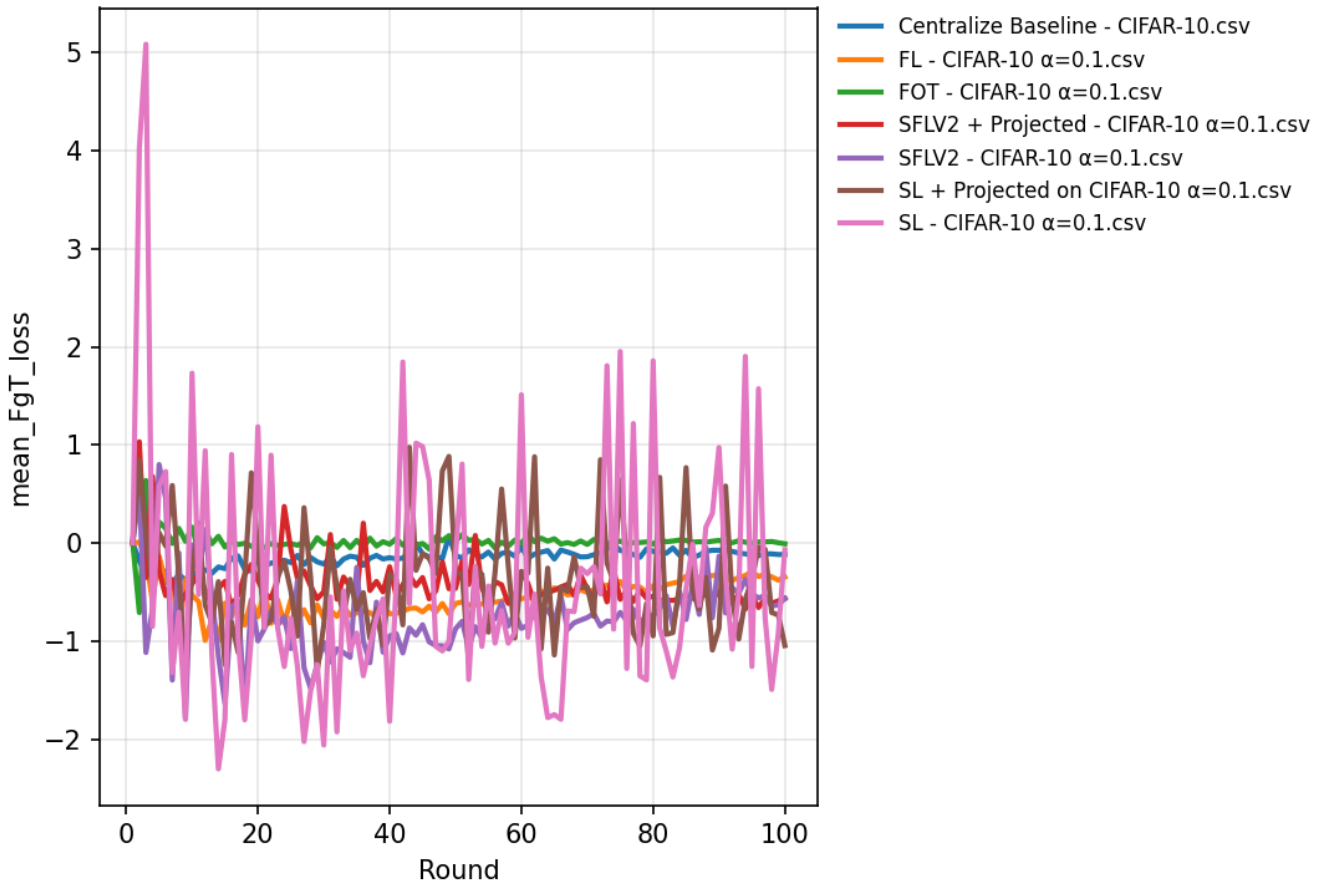

Figure 6: Quantifying Forgetting Comparison between Tested Models